

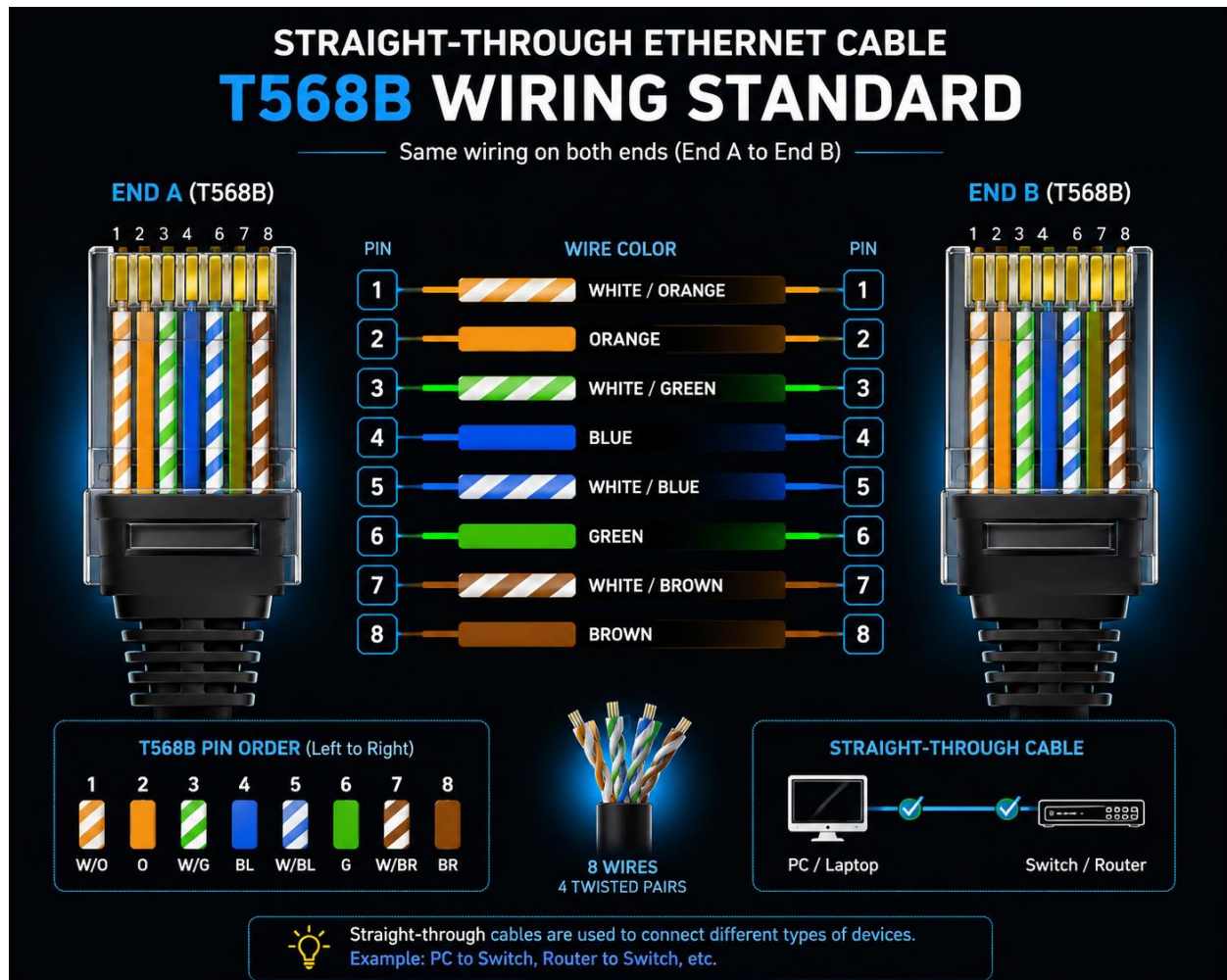
1 List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

Difference	Ethernet Cable	Fiber Optic Cable
1. Transmission Medium	Uses electrical signals through copper wires.	Uses light signals through glass or plastic fibers.
2. Speed & Distance	Suitable for normal network speeds and shorter distances.	Supports very high speeds and much longer distances.
3. Interference	Can be affected by electromagnetic interference (EMI).	Not affected by electromagnetic interference, so it provides more reliable communication.

Real-World Examples:

1. **Ethernet Cable:** Commonly used to connect a **home Wi-Fi router to a computer, printer, or network switch.**
2. **Fiber Optic Cable:** Commonly used as the **backbone connection between buildings on a college campus or inside a data center,** where high speed and long-distance communication are required.

2 Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.



Explanation:

A **straight-through Ethernet cable** is a network cable in which **both ends use the same T568B wiring standard**. It has **8 wires arranged in the same pin order** on both RJ45 connectors.

The T568B pin order is:

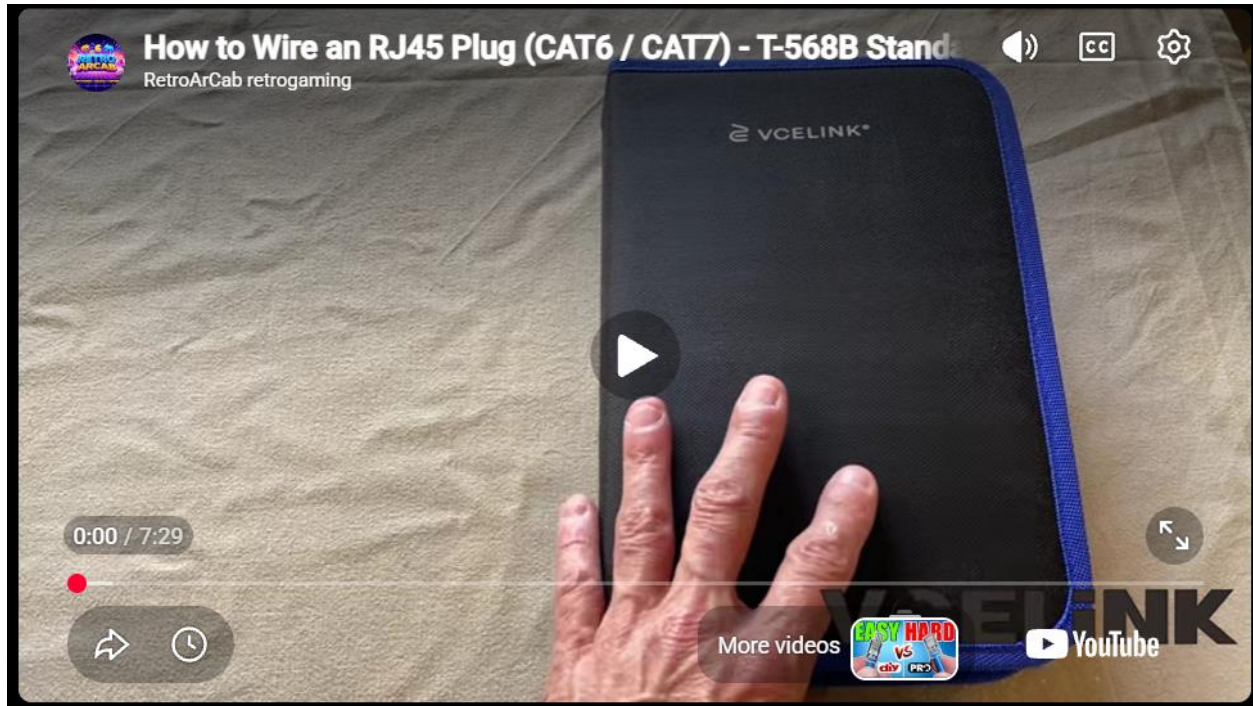
1. **White/Orange**
2. **Orange**
3. **White/Green**
4. **Blue**
5. **White/Blue**
6. **Green**
7. **White/Brown**
8. **Brown**

Because the wiring is the **same on both ends**, it is called a **straight-through cable**. It is commonly used to connect **different types of network devices**, such as a **computer to a switch** or a **router to a switch**.

➤ **In short:**

End A (T568B) → Same wire order → End B (T568B).

3 Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.



1. Prepare the tools:

Take a CAT6 cable, RJ45 connector, crimping tool, cable cutter/stripper, and a LAN cable tester.

2. Cut the cable:

Cut the CAT6 cable to the required length.

3. Remove the outer cover:

Use the cable stripper to remove about **2–3 cm** of the outer cover. Be careful not to damage the wires inside.

4. Separate the wires:

Separate the **8 wires** inside the cable and straighten them carefully.

5. Arrange the wires in T568B order:

Arrange the wires from **Pin 1 to Pin 8** as follows

- 6. Pin 1 – White/Orange
- 7. Pin 2 – Orange
- 8. Pin 3 – White/Green
- 9. Pin 4 – Blue
- 10. Pin 5 – White/Blue
- 11. Pin 6 – Green
- 12. Pin 7 – White/Brown
- 13. Pin 8 – Brown

14. Cut the wires evenly:

Make all 8 wires equal in length and keep their order the same.

15. Insert the wires into the RJ45 connector:

Carefully insert all 8 wires into the RJ45 connector. Make sure they reach the front of the connector and remain in the correct T568B order.

16. Check the wires:

Before crimping, check that all wires are in the correct position and the cable cover is properly inside the connector.

17. Crimp the connector:

Put the RJ45 connector into the **crimping tool** and press it firmly. This locks the wires into the connector.

18. Test the cable:

Use a **LAN cable tester** to check the cable. If the pins light up in the correct order, the cable has been connected correctly.

4 Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.

Hint: Consider speed, distance, and budget for your answer.

SETTING UP A GAMING CAFÉ

10 PCS FOR LAN GAMING – BEST CHOICE

CABLE TYPE

ETHERNET (CAT6)



Best for short distance, high speed & budget

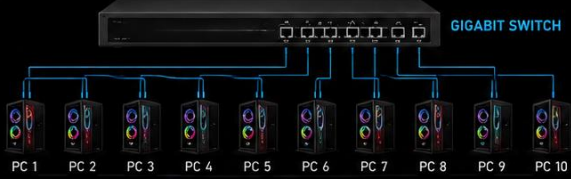
TOOL

RJ45 CRIMPER



Easy to use, reliable & cost effective

NETWORK SETUP (10 PCS)



GIGABIT SWITCH

PC 1 PC 2 PC 3 PC 4 PC 5 PC 6 PC 7 PC 8 PC 9 PC 10

WHY THIS CHOICE?

- ✓ 10 PCs are in a small area, so distance is short.
- ✓ CAT6 Ethernet provides more than enough speed for smooth LAN gaming.
- ✓ It is budget friendly and easy to install.
- ✓ RJ45 crimper is simple to use and widely available.

ETHERNET (CAT6) CONNECTION



PC CAT6 CABLE (Up to 100 m) SWITCH

WHY ETHERNET (CAT6)?

- SPEED**
Up to 1 Gbps
Perfect for LAN gaming
- DISTANCE**
Works great for short distances (up to 100 m)
- BUDGET**
Affordable cable and tools, low cost setup

WHY NOT FIBER OPTIC?



- ✗ More expensive (cable, tools & equipment)
- ✗ Requires a fiber cleaver and special skills
- ✗ Ideal for long distances (hundreds of meters), not needed for a small gaming café

FINAL CHOICE

ETHERNET (CAT6) + RJ45 CRIMPER

BEST SPEED, EASY SETUP AND BUDGET FRIENDLY!

- For a gaming café with **10 PCs**, I would use **Ethernet (CAT6) cables** because they provide good speed for LAN gaming and work well over short distances.
- I would use an **RJ45 crimper** to connect the RJ45 connectors to the Ethernet cables.
- Ethernet cables are **cheaper and easier to install** than fiber optic cables.
- Fiber optics and a fiber cleaver would be unnecessary and more expensive for a small gaming café.